Ceftriaxone

Mechanism of action

- Ceftriaxone is a cephalosporin antibiotic with broad activity against gram-negative and gram-positive bacteria. It inhibits production of the bacterial cell wall, causing bacteria to die.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Suspected meningococcal septicaemia.
- ✓ Sepsis, where cefazolin is not indicated, and:
  - The patient is aged greater than or equal to 12 years, and
  - One or more clinical features indicating antibiotics are present, and
  - Time to hospital is greater than 30 minutes.

Contraindications

- ✕ Anaphylaxis to cephalosporins.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- 2 g IV for an adult.
- 2 g IM for an adult if IV access cannot be immediately obtained (meningococcal septicaemia only).
- See the paediatric drug dose tables for a child.

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Administration

- IV administration:
  - Add approximately 4 ml of 0.9% sodium chloride to a 2 g ampoule and shake until dissolved.
  - Draw up the ampoule and dilute to a total of 10 ml.
  - Discard unrequired volume before administration for a child.
  - Administer IV over 1-2 minutes preferably into a running line.
- IM administration:
  - Add 4 ml of 0.9% sodium chloride to a 2 g ampoule and shake until dissolved.
  - Draw up the ampoule using two syringes with approximately half in each. The total volume will be 5 ml.
  - Discard unrequired volume before administration for a child.
  - Administer one syringe into each lateral thigh. If this site is not suitable, use each lateral upper arm.

Common adverse effects

- None.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 24 hours.

Usual preparation

- Ampoule containing 2 g of ceftriaxone as a powder for reconstitution.

Pharmacokinetics

- 50% is excreted in urine and 50% in bile.
- Neither renal impairment nor hepatic impairment alter the initial (loading) dose.

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Common interactions

- None.

Additional information

- Some references describe dissolving ceftriaxone using 1% lignocaine for IM injection, to reduce the pain of injection. In the setting of suspected meningococcal septicaemia this reduction in pain is not significant and ceftriaxone should be dissolved using 0.9% sodium chloride.
- As meningococcal bacteria die they release endotoxins. The body's immune response to endotoxins can cause profound worsening of shock following antibiotic administration. Be prepared to treat this with 0.9% sodium chloride IV, and metaraminol or adrenaline IV. It is rare for significant amounts of endotoxins to be released from other bacteria.

More Information

Version history

11 September 2023 1.0.4
Update: Indications updated.